

Intro

Agents can work together in a variety of ways to solve problems. Research works like [AutoGen](#), [MetaGPT](#) and [ChatDev](#) have shown multi-agent systems out-performing single agent systems at complex tasks like software development.

A multi-agent design pattern is a structure that emerges from message protocols: it describes how agents interact with each other to solve problems. For example, the [tool-equipped agent](#) in the previous section employs a design pattern called ReAct, which involves an agent interacting with tools.

You can implement any multi-agent design pattern using AutoGen agents. In the next two sections, we will discuss two common design patterns: group chat for task decomposition, and reflection for robustness.